FINAL PROJECT · PRE-STUDY

Seven Capstone Projects

What each project does, and what it gains over the textbook default

Advanced Financial Time Series · Units 5–16

Read this before choosing a project. Every chart here uses simulated data, so the patterns are clean and the true values are known.

Each project upgrades one textbook habit

Three ideas to carry into the final project

1

Every project upgrades a default you already know

Constant volatility, a fixed hedge ratio, a single regression: each project shows what the habit misses.

2

Honesty decides the grade

Fit on 2015–2021, score on 2022–2026. Losing honestly to a naive benchmark can still earn full marks.

3

Claude Code does the typing; you do the judging

It downloads, codes, runs and fixes. You choose the question, check every number and explain it.

Choose the project whose upgrade you most want to understand.

Traditional methods rest on five hidden assumptions

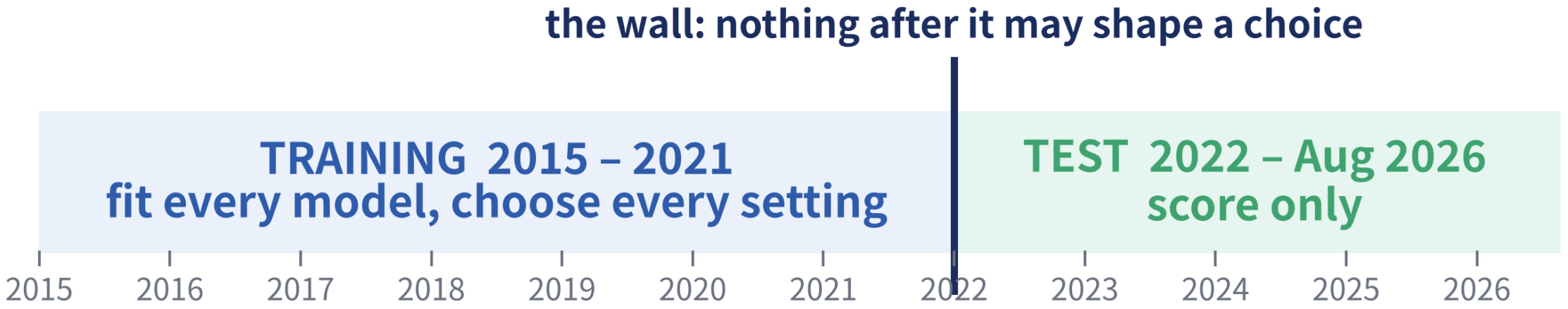
Background: what "traditional" means in this deck

Traditional habit	Hidden assumption	Fixed in
Constant volatility	Today's risk equals the average risk of the past	Projects 1, 7
One fixed coefficient	A relationship never drifts	Projects 2, 6
One regime	Markets behave the same in calm and crisis	Project 3
One variable, one frequency	Other series and faster data add nothing	Projects 4, 5
One point estimate	The estimate carries no uncertainty	Projects 6, 7

Each habit is a sensible first model. The projects ask when it breaks and what that costs.

Every project follows the same honest design

One wall separates fitting from scoring; the five rules build on it



1	2	3	4	5
One question	Fixed split	Fit on training	Beat the benchmark	Grade on test only

A benchmark and a leak check judge every result

Definitions used on every project slide

Naive benchmark

- The simplest forecast a sceptic would accept: random walk, historical average, buy-and-hold.
- If a model cannot beat it out of sample, its extra complexity has not paid off.
- Losing to it is a valid finding when reported honestly.

Look-ahead bias (leakage)

- Using information that did not exist on the decision date.
- Common leaks: full-sample scaling, smoothed estimates, thresholds tuned on the test period.
- It makes any model look brilliant, so it carries the most points (20 of 100).

Claude Code builds; you decide and verify

The same five prompts open every project

[Input] Prompts · Claude Code (full text in the guide)

```
1 Setup   Clone the course repo, pull data to
          2026-08-20, show me the manifest.

2 Rules   Write CLAUDE.md: question, split, benchmark,
          no test data in any choice.

3 Build   Paste the project's starter prompt.

4 Audit   Find every look-ahead leak, fix it,
          re-run, show what changed.

5 Report  Draft report.md from results/ only,
          then build the PDF.
```

Your job

- Choose the question and the design
- Check that every number is plausible
- Write the conclusion in your own words
- Be ready to explain any line live

Seven projects, each built around one upgrade

Choose by the habit you want to break, not only by difficulty

No.	Project	Upgrade over	Level
1	Volatility contest and VaR backtest	Constant-volatility VaR	★★
2	Pairs trading with a Kalman hedge ratio	One fixed OLS hedge ratio	★★★
3	Regime switching with an HMM	Buy-and-hold, one regime	★★
4	Nowcasting US GDP	Waiting for GDP; AR on past GDP	★★★
5	US-to-Korea shocks with a VAR	Correlation, single regressions	★★
6	Time-varying beta with the Kalman filter	Rolling-window OLS beta	★★★
7	MCMC samplers on stochastic volatility	GARCH point estimate; one sampler	★★★

Volatility forecasts must react when risk changes

Question: which model gives the most reliable 99% VaR for the S&P 500?

$$\sigma_t^2 = \omega + \left(\alpha + \gamma\, \mathbf{1}\{r_{t-1} < 0\}\right) r_{t-1}^2 + \beta\, \sigma_{t-1}^2, \qquad \widehat{\mathrm{VaR}}_t = -z_{0.01}\, \hat{\sigma}_t$$

Traditional: constant volatility

- One σ from the past, or the 1% quantile of the last 250 days
- Reacts slowly when a crisis starts
- Breaches arrive in clusters, exactly when it hurts

This project: GARCH + backtests

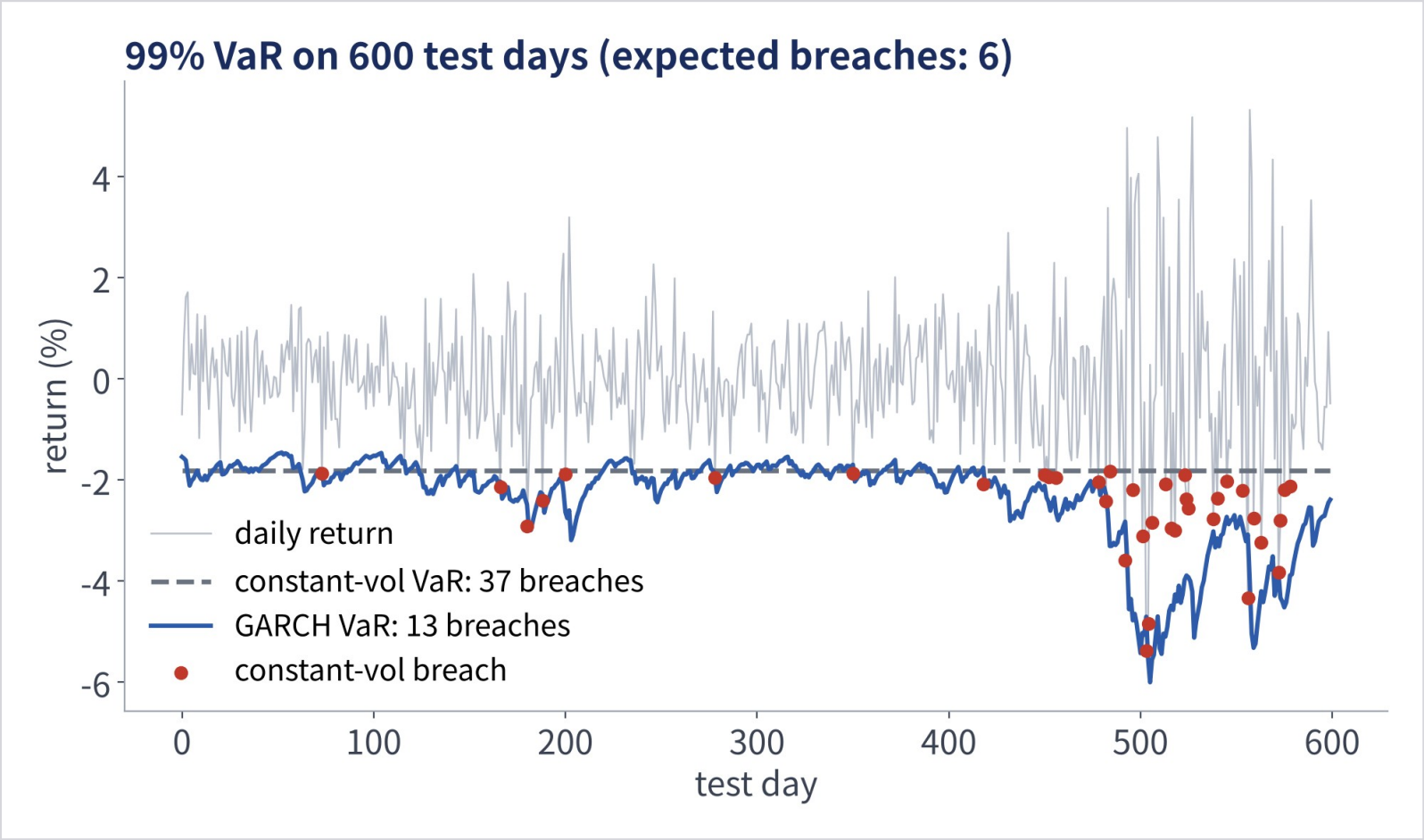
- σ updates every day from yesterday's shock
- GJR adds the bad-news effect; Student-t adds fat tails
- Kupiec and Christoffersen tests check count and clustering

A volatility model cuts VaR breaches in a storm

Simulated GARCH returns; both VaR lines are fitted on training data only

What to see

- Expected breaches: 6
- Constant σ: 37, bunched in the storm
- GARCH: 13; its line drops as risk rises
- Still too many: test, never assume



A drifting hedge ratio needs daily re-estimation

Question: does a Kalman hedge ratio trade KO–PEP better after costs?

$$y_t = a_t + \gamma_t\, x_t + \varepsilon_t, \qquad \begin{pmatrix} a_t \\ \gamma_t \end{pmatrix} = \begin{pmatrix} a_{t-1} \\ \gamma_{t-1} \end{pmatrix} + w_t, \qquad \text{half-life} = \frac{\ln 2}{\theta}$$

Traditional: one OLS hedge ratio

- Estimated once on the training period
- If the relationship drifts, the spread trends away
- Signals fire and never revert: losses, not profits

This project: Kalman + OU timing

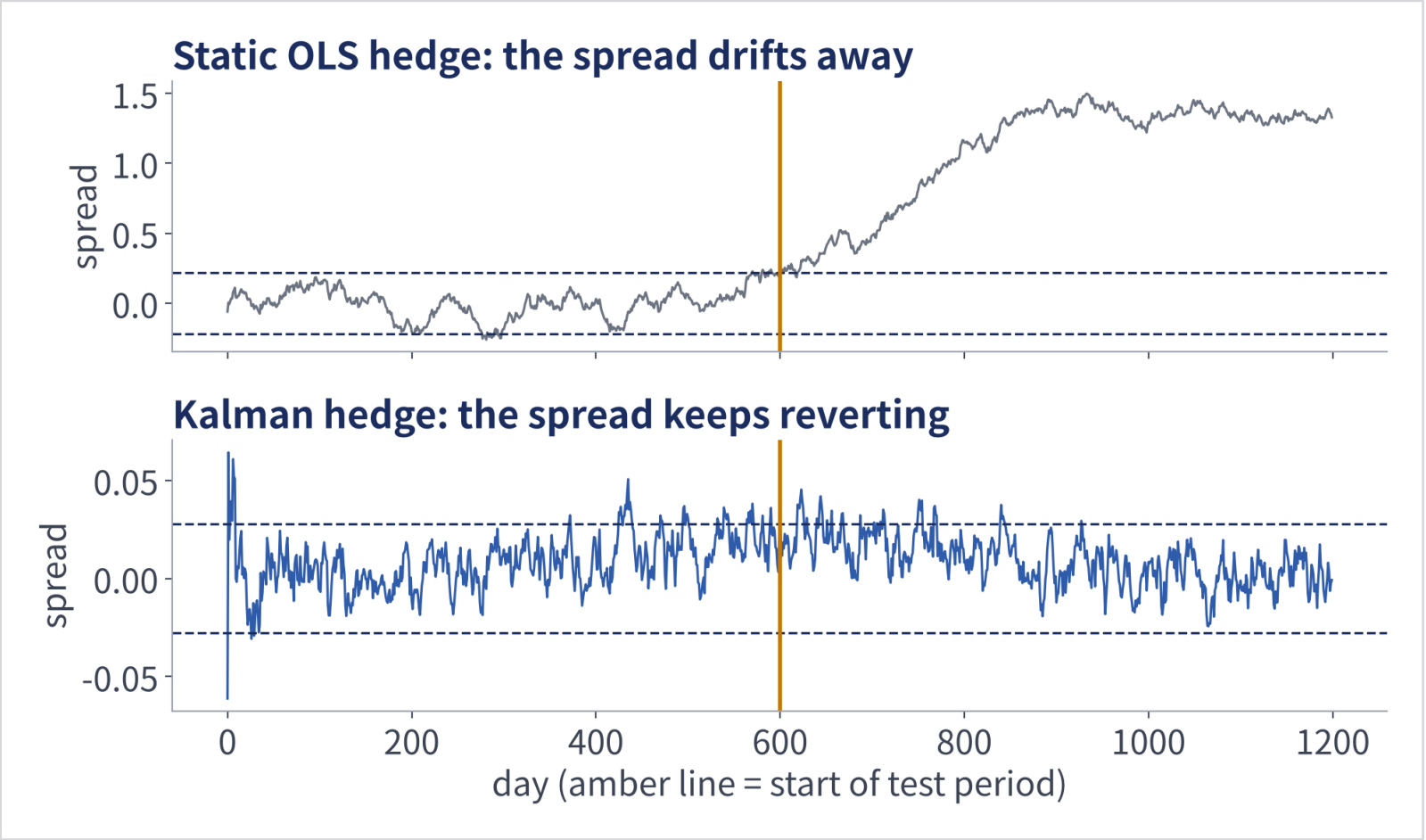
- Intercept and slope update with each new day
- The OU half-life sets how long to hold
- Costs from 0 to 20 bp show whether the edge survives

A static hedge turns a reverting spread into a trend

Simulated pair whose true hedge ratio rises from 0.8 to 1.3

What to see

- Top: the static spread leaves its ±2σ band
- Bottom: the Kalman spread keeps reverting
- Trade only the past-only (filtered) estimate



Markets switch between calm and turbulent regimes

Question: can regime probabilities beat buy-and-hold after risk?

$$P(S_t = j \mid S_{t-1} = i) = p_{ij}, \qquad r_t \mid S_t = k \sim N(\mu_k, \sigma_k^2), \qquad \xi_t = P(S_t = \text{calm} \mid r_1, \ldots, r_t)$$

Traditional: one model for every day

- One mean and one volatility for all days
- Or ad hoc rules such as moving-average crossings
- No measure of how sure we are about the state

This project: two-state HMM

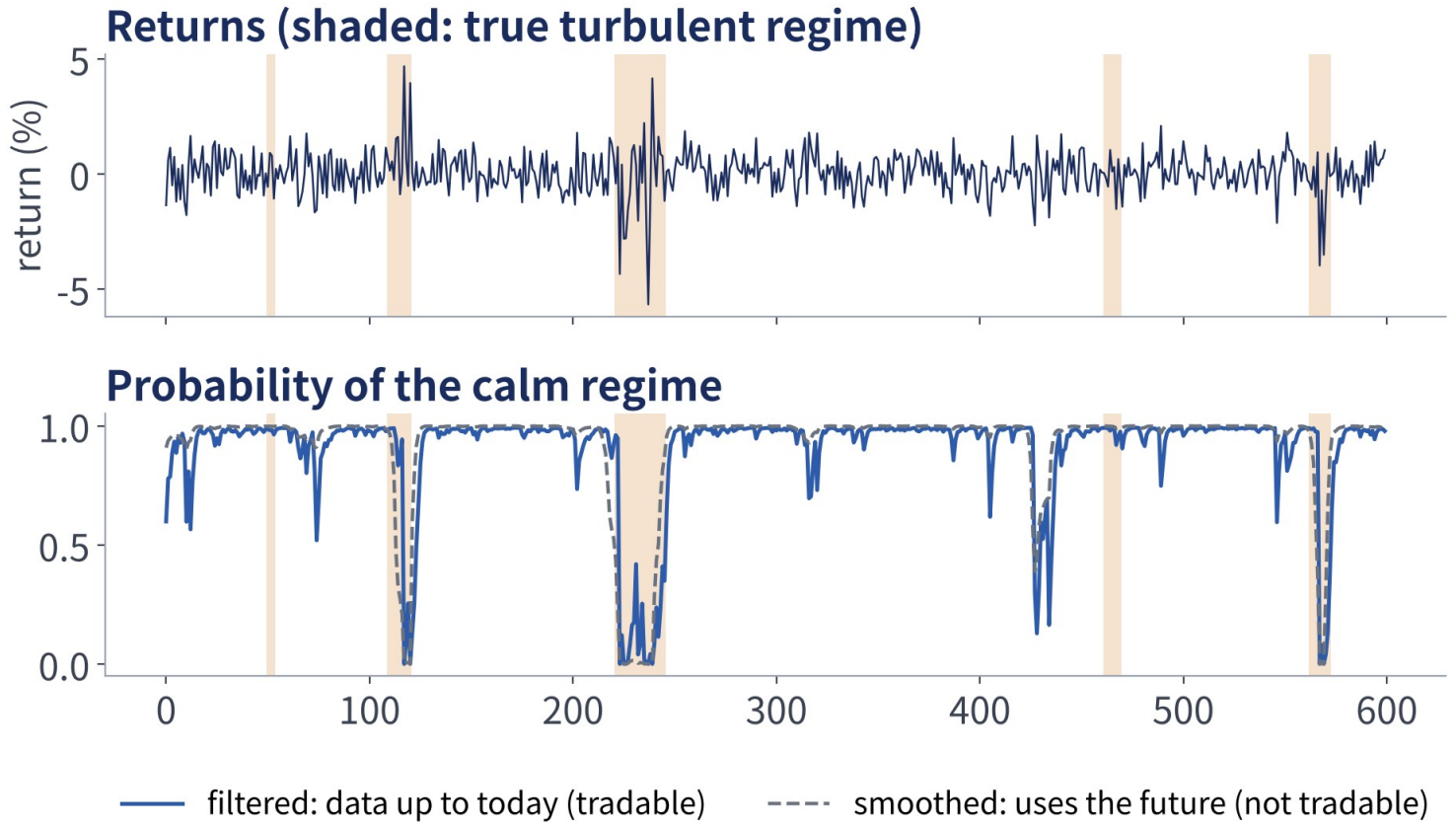
- Each regime has its own mean, volatility and persistence
- Gives a probability, not a hard label
- The filtered probability uses only past data

Only the filtered probability can be traded

Simulated two-regime returns; the true turbulent regime is shaded

What to see

- Both curves find the storms
- Smoothed (dashed) reacts early: it has seen the future
- Trading on it inflates the backtest



Monthly data reveal GDP growth before it is published

Question: when does a factor nowcast start to beat an AR(1) on past GDP?

$$x_{i,t} = \lambda_i f_t + e_{i,t}, \qquad f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + u_t, \qquad z_t = W_t (Z\alpha_t + \varepsilon_t)$$

Traditional: wait, or extrapolate

- GDP arrives about a month after the quarter ends
- An AR model uses past GDP only
- Bridge equations use one indicator at a time

This project: dynamic factor model

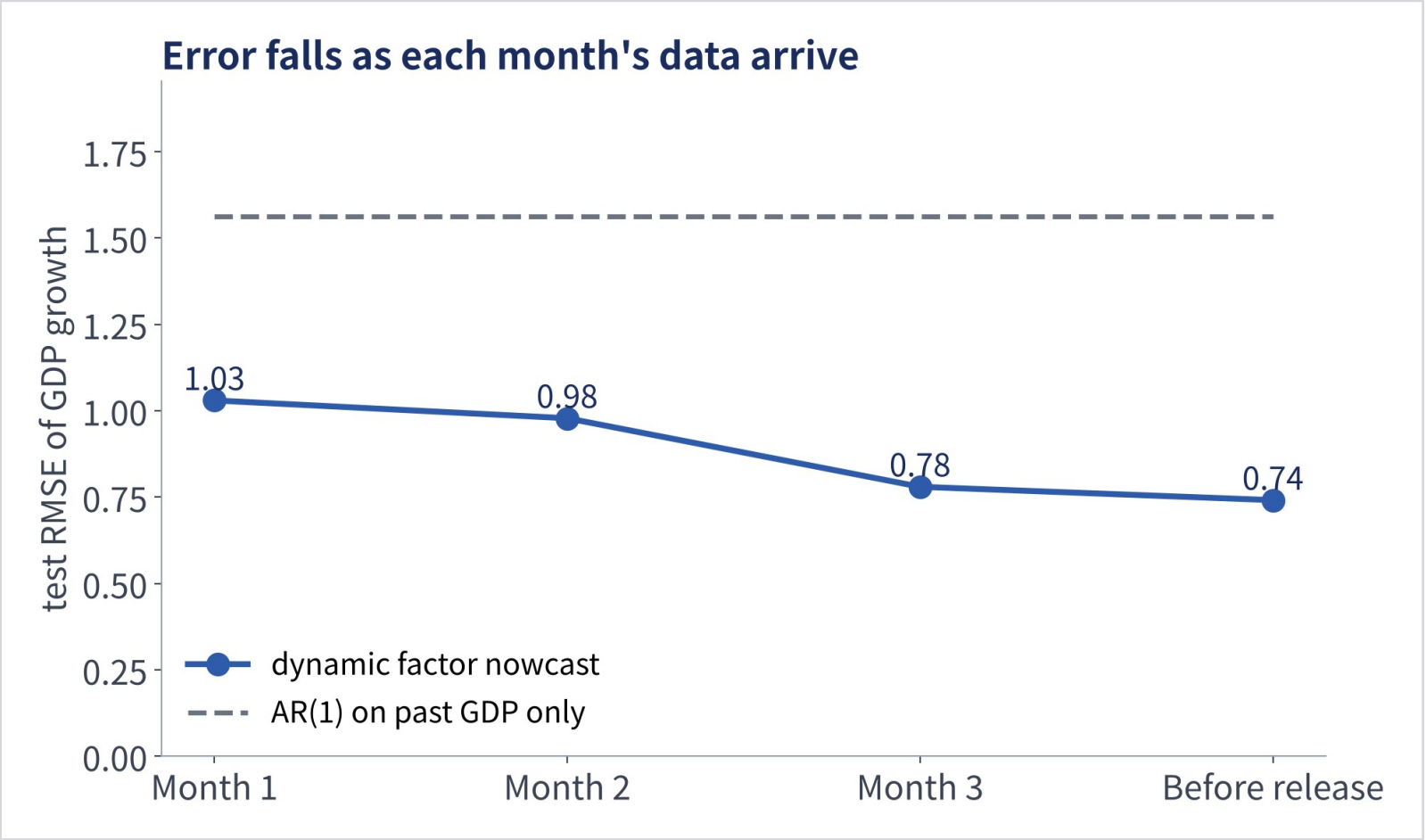
- Five monthly indicators share one common factor
- The Kalman filter treats unpublished values as missing
- The nowcast updates with every release

Nowcast error falls as each month's data arrive

Simulated factor data with publication lags; RMSE on test quarters

What to see

- Dashed AR(1) never learns within the quarter
- The factor nowcast improves month by month
- The size of the gain is your answer



Shocks travel between markets over several days

Question: do yesterday's US return and VIX change forecast today's KOSPI?

$$y_t = c + A_1\, y_{t-1} + \cdots + A_p\, y_{t-p} + u_t, \qquad \mathrm{IRF}(h) = \frac{\partial\, y_{t+h}}{\partial\, \varepsilon_t}$$

Traditional: one equation

- Same-day correlation mixes cause and timing
- One equation ignores feedback between markets
- No picture of how long a shock lasts

This project: VAR, aligned clocks

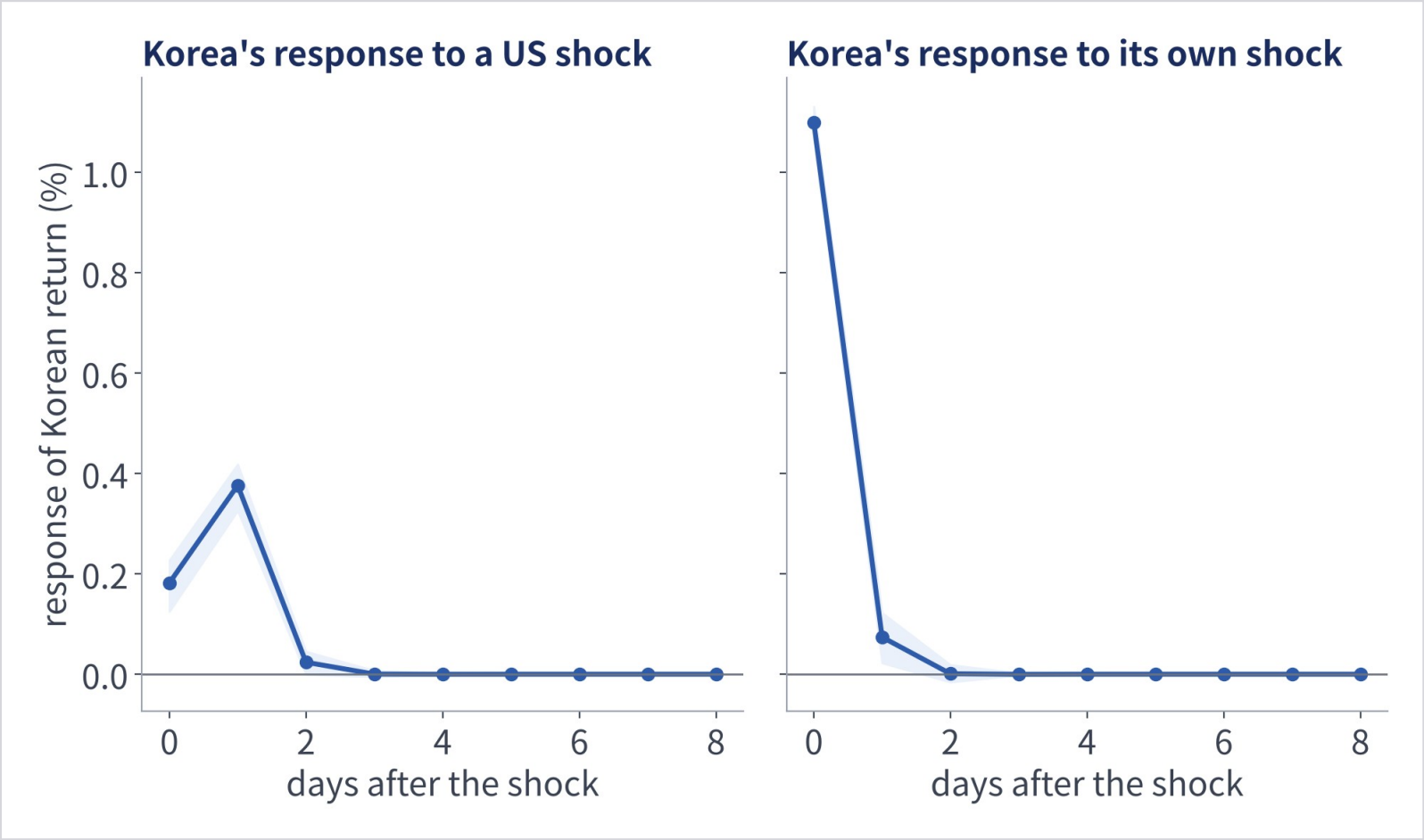
- All series modelled jointly with their lags
- Granger tests ask who predicts whom
- Impulse responses trace a shock day by day

An impulse response shows size and duration

Simulated US and Korean returns; bands from 300 bootstrap runs

What to see

- A US shock moves Korea most the next day
- The effect is gone within two days
- Korea's own shock hits at once, then fades



A Kalman beta adapts at a data-driven speed

Question: does a Kalman beta hedge AAPL better than a rolling OLS beta?

$$y_t = \beta_t\, x_t + \varepsilon_t, \quad \beta_t = \beta_{t-1} + \eta_t, \qquad m_t = m_{t-1} + K_t\,(y_t - x_t\, m_{t-1})$$

Traditional: rolling-window OLS

- The window length is an arbitrary choice
- Short window: noisy; long window: slow
- Old days drop out abruptly, all weights equal

This project: Kalman beta

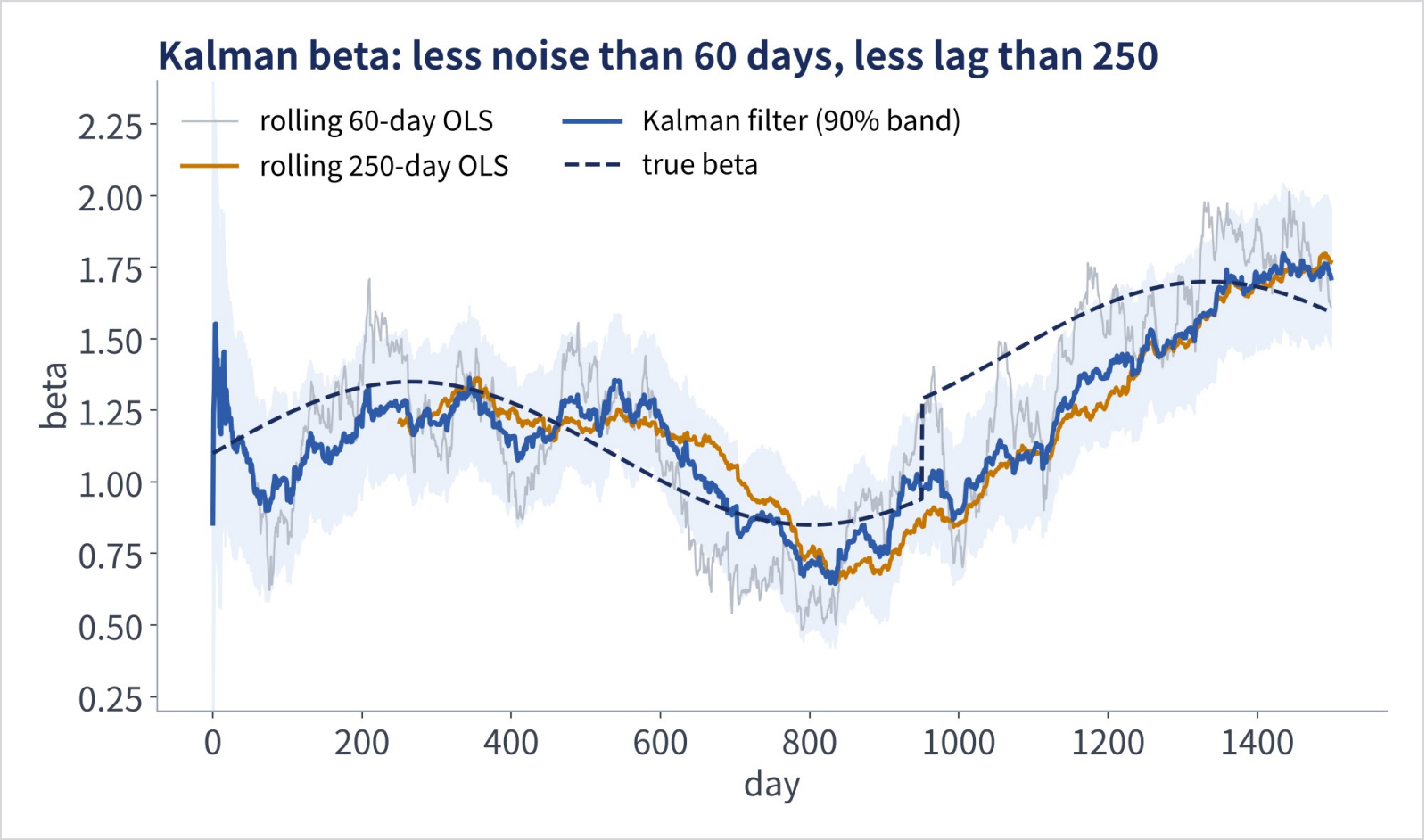
- The gain K sets how far each day moves the beta
- The likelihood chooses the speed q = W/V
- A band of $\pm 1.645\sqrt{C}$ shows the uncertainty

A Kalman beta balances noise against lag

Simulated returns whose true beta drifts and jumps at day 950

What to see

- Grey 60-day beta: noisy
- Amber 250-day beta: late at the jump
- Blue Kalman: in between, with a band
- Judge it by the hedged variance



Stochastic volatility calls for MCMC sampling

Question: which MCMC method can be trusted with one hidden state per day?

$$r_t = e^{h_t/2}\,\varepsilon_t, \quad h_t = \mu + \phi\,(h_{t-1} - \mu) + \sigma_\eta\,\eta_t, \qquad \mathrm{ESS} = \frac{S}{1 + 2\sum_{k\geq 1}\rho_k}$$

Traditional: one fit, one sampler

- GARCH by maximum likelihood: a point estimate
- Run one sampler once and believe the output
- No check that the chain has converged

This project: three samplers

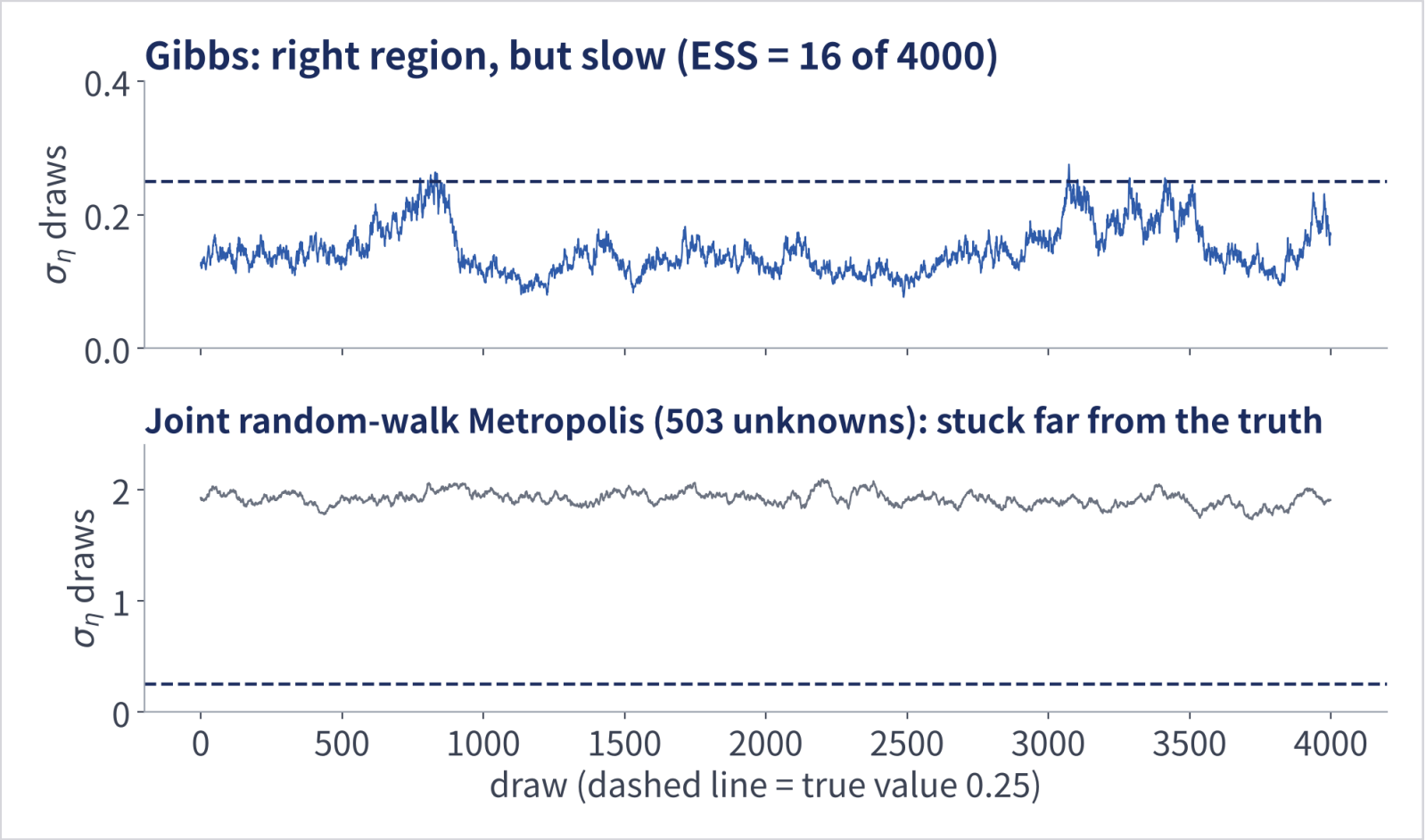
- Volatility gets its own shock: a hidden state
- A full posterior, not just a point
- ESS, R-hat and divergences judge each sampler

A sampler can look stable and still be wrong

Simulated SV over 500 days; true volatility of volatility is 0.25

What to see

- Gibbs: right region, slow mixing
- Joint Metropolis: stuck far away
- One chain cannot reveal this
- Run four chains and check R-hat



Choose the question you want answered

Four questions to narrow the choice

Q1	Do you want to trade, and see whether costs eat the profit?	**Projects 2, 3**
Q2	Do you want to measure and forecast risk?	**Projects 1, 6, 7**
Q3	Do you prefer macro data and links between markets?	**Projects 4, 5**
Q4	Do you want a ★★ start with room to extend?	**Projects 1, 3, 5**

Three steps start the project well

Set in week 15, due in week 16

1 **Read your project's section in the guide**

Question, data, models, benchmark and the starter prompt are all written out.

2 **Run Prompts 1 and 2 on day one**

Pull the data and write CLAUDE.md, so the split is fixed before any model sees the data.

3 **Keep prompts.md from the start**

It records how you worked and makes the oral check easy.

Graded on honesty first: an honest loss beats a leaky win.